King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 101 Major Exam I Term 242

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. The equation of the tangent line to the curve $f(x) = x^2 + 2x - 1$ at the point $(1, 2)$ is:

(a) $y = 2x - 4$

(b) $y = 2x - 1$

(c) $y = 4x - 2$

(d) $y = 4x + 2$

(e) $y = 2x + 4$

2. The function $f(x) = \begin{cases} \ln(x+1), & x \geq 0 \\ 1 - x^2, & x < 0 \end{cases}$

(a) has a removable discontinuity at $x = 0$

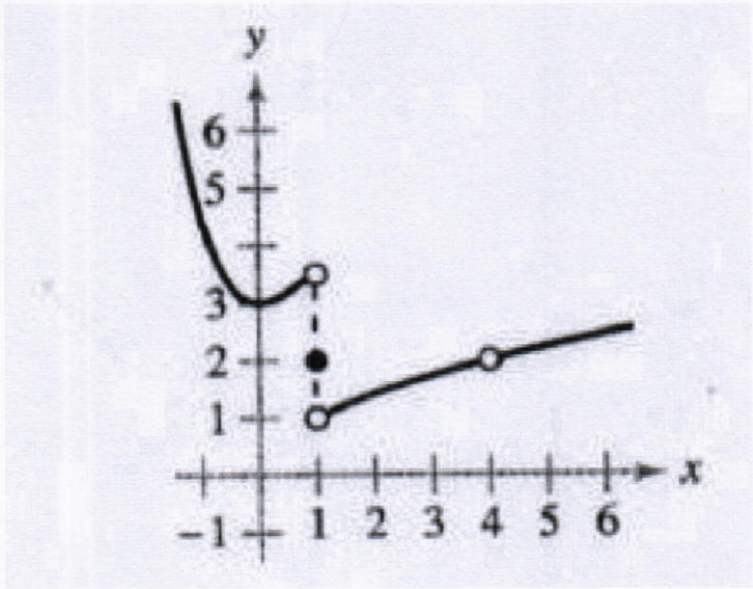
(b) is continuous on $(-\infty, \infty)$

(c) has a removable discontinuity at $x = 1$

(d) has a non removable discontinuity at $x = 1$

(e) has a non removable discontinuity at $x = 0$

3. One statement is TRUE about the given graph



(a) $3 < \lim_{x\to 1} f(x) < 4$

(b) $\lim_{x\to 4} f(x) = 2$

(c) $\lim_{x\to 4} f(x)$ does not exist

(d) $\lim_{x\to 1} f(x) = 2$

(e) $\lim_{x\to 1} f(x) = 1$

4. $\lim_{x\to\infty} \frac{5x^3+1}{10x^3-3x^2+7} =$

(a) 2

(b) $\frac{1}{2}$

(c) $\frac{1}{10}$

(d) $\frac{5}{7}$

(e) $\frac{1}{7}$

5. $\lim_{x\to -2^+} \frac{1}{x+2}$

(a) goes to $-\frac{1}{4}$

(b) goes to $-\infty$

(c) goes to zero

(d) goes to $\frac{1}{4}$

(e) goes to $+\infty$

6. The function $f(x) = \dfrac{x}{x^2 - 9}$

(a) has one vertical asymptote, moreover $\lim_{x \to 3^+} f(x) = -\infty$

(b) has 2 vertical asymptotes, moreover $\lim_{x \to 3^-} f(x) = -\infty$

(c) has 2 vertical asymptotes, moreover $\lim_{x \to 3^-} f(x) = +\infty$

(d) has no vertical asymptote

(e) has one vertical asymptote, moreover $\lim_{x \to 3} f(x) = +\infty$

7. $\lim_{x \to \infty} x \sin\left(\dfrac{1}{x}\right) =$

(a) $-\infty$

(b) 1

(c) 0

(d) -1

(e) ∞

8. If $f(x) = \dfrac{3}{(2x)^3} + 2\sin x$, then $f'\left(\dfrac{\pi}{2}\right)$ is equal to

(a) $\dfrac{-27}{\pi^4}$

(b) $\dfrac{-72}{\pi^4}$

(c) $\dfrac{-18}{\pi^4}$

(d) $\dfrac{-9}{\pi^4}$

(e) $\dfrac{-36}{\pi^4}$

9. The domain of $f(x) = \dfrac{\sqrt{x}}{|x| - 4}$ is

(a) $[0, 4) \cup (4, \infty)$

(b) $(0, \infty) - \{4\}$

(c) $(-\infty, \infty) - \{\pm 4\}$

(d) $(0, \infty)$

(e) $(-4, 0) \cup (0, 4)$

10. $\lim_{x\to 0} \frac{e^{2x}-1}{e^x-1}$

(a) does not exist

(b) goes to $-\infty$

(c) equals zero

(d) equals 2

(e) equals 1

11. $\lim_{x\to\infty} (e^x + e^{-x} + e) =$

(a) e

(b) ∞

(c) $-\infty$

(d) $e + \frac{1}{e}$

(e) $2 + e$

12. One statement is false about $f(x) = \frac{\sin x}{x}$

(a) $f(x)$ is an even function

(b) $x = 0$ is a vertical asymptote

(c) Domain of $f(x)$ is $(-\infty, 0) \cup (0, \infty)$

(d) $\lim_{x\to-\infty} f(x) = 0$

(e) $\lim_{x\to 0^+} f(x) = 1$

13. $\lim_{x\to 4} \frac{\sqrt{x+5}-3}{x-4} =$

(a) $\frac{1}{7}$

(b) $\frac{1}{5}$

(c) $\frac{1}{3}$

(d) $\frac{1}{4}$

(e) $\frac{1}{6}$

14. Question 11 of the original exam is omitted (marked as omitted on the original exam).

Answer Key

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Q	Ans	Brief explanation
1	**(c)**	$f'(1) = 4$: $y - 2 = 4(x - 1)$, so $y = 4x - 2$.
2	**(e)**	At $x = 0$: left limit 1, right limit 0, a jump (non removable).
3	**(b)**	Both sides approach height 2 at $x = 4$; at $x = 1$ the one-sided limits differ.
4	**(b)**	Equal degrees: ratio of leading coefficients $\frac{5}{10} = \frac{1}{2}$.
5	**(e)**	$x + 2 \to 0^+$, so $\frac{1}{x+2} \to +\infty$.
6	**(b)**	Zeros of denominator ± 3; just left of 3, $f < 0$, so $-\infty$.
7	**(b)**	$t = 1/x \to 0$: $\frac{\sin t}{t} \to 1$.
8	**(c)**	$f' = -\frac{9}{8}x^{-4} + 2\cos x$, so $f'(\pi/2) = -\frac{18}{\pi^4}$.
9	**(a)**	$x \geq 0$ and $x \neq 4$: $[0, 4) \cup (4, \infty)$.
10	**(d)**	$\frac{(e^x-1)(e^x+1)}{e^x-1} = e^x + 1 \to 2$.
11	**(b)**	$e^x \to \infty$, $e^{-x} \to 0$: the sum goes to ∞.
12	**(b)**	$\frac{\sin x}{x} \to 1$ at 0, so no vertical asymptote there.
13	**(e)**	Conjugate: $\frac{1}{\sqrt{x+5}+3} \to \frac{1}{6}$.
14	omitted	Omitted.